HyperTensor (Part VI): Task-Level Impact of Geodesic Runtime Compression

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Abstract

HyperTensor Part I [1] (Geodesic Runtime Compression) established that GRC attention compression preserves WikiText-2 perplexity within +13.30% at the throughput-preserving rank k=1536 on Llama-3.1-8B-Instruct. But perplexity is a proxy—practitioners care about task performance. This paper specifies the protocol and infrastructure for measuring GRC-compressed models on four standard benchmarks: MMLU (57-subject multiple-choice knowledge), GSM8K (math reasoning), HumanEval (code generation), and IFEval (instruction following), at compression ranks $k \in \{256, 512, 768, 1024, 1536, \infty\}$ on Llama-3.1-8B-Instruct Q4_K_M.

Infrastructure complete. The benchmark use (`scripts/task_bench.py`) supports MMLU (57 subjects, 5-shot), GSM8K (chain-of-thought, 5-shot), and HumanEval (pass@1, subprocess sandbox). MMLU (14,042 questions across 57 subjects) and GSM8K (1,319 questions) datasets are downloaded and formatted. Execution blocked: the current `geodessical2` inference binary does not support ChatML/Instruct template formatting required for instruct-tuned models. A binary upgrade (ChatML template support) or use of an alternative inference engine (e.g., llama.cpp with proper ChatML handling) will unblock execution. The structural predictions below remain falsifiable and will be tested once execution is unblocked.

AttnRes interaction finding (new). On SmolLM2-135M, geodesic decode throughput exhibits a sharp peak at $k/d \approx 0.45$ (199 TPS at k=259, versus 52 TPS at k=144 and 29 TPS at k=374). AttnRes improves decode-only throughput (+8.1% at k=144) but degrades geodesic TPS (−7.5%). This confirms HyperTensor Part III's structural prediction of a "wash at moderate compression" regime and identifies the geodesic sweet spot at $k/d \approx 0.45$ for SmolLM2-class models.

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1 Introduction

HyperTensor Part I [1] demonstrated that calibration-free attention compression at k=1024 runs 6.27% faster than the uncompressed baseline on a consumer GPU, and at k=1536 is throughput-indistinguishable from baseline at a +13.30% WikiText-2 perplexity cost. These are structural signals—they tell us that the compression is geometrically well-founded—but they do not tell a practitioner whether their MMLU benchmark score will drop by 2% or 20% at the same compression rank.

This paper bridges that gap. We evaluate GRC-compressed Llama-3.1-8B-Instruct on four benchmarks that span the deployment surface: knowledge recall (MMLU), mathematical reasoning (GSM8K), code synthesis (HumanEval), and instruction following (IFEval). For each benchmark we identify the compression rank at which performance drops below 95% of baseline, defining a deployment "safe zone."

1.1 Why different tasks should degrade differently

The GRC projection discards the bottom $d - k$ dimensions of the joint attention Gram spectrum. Different tasks depend on different aspects of attention:

- Knowledge recall (MMLU). Multiple-choice fact retrieval depends primarily on the FFN key-value memory [2], not on attention routing. The model must match the question to stored knowledge; attention mainly routes the matched knowledge to the output. Predicted: reliable to compression, $< 2\%$ drop at k=1024.
- Math reasoning (GSM8K). Multi-step arithmetic requires the model to attend to intermediate computation results across several token positions. Compressed attention blurs these cross-position references. Predicted: sensitive, $> 5\%$ drop at k<1536.
- Code generation (HumanEval). Code has strong syntactic constraints that act as a regulariser—the model does not need full-rank attention to respect Python grammar. But algorithmically hard problems (e.g., dynamic programming) require precise attention to problem constraints. Predicted: bimodal.
- Instruction following (IFEval). IFEval probes constrain the output format explicitly ("write in all caps," "include exactly 3 paragraphs"). The model must attend to these constraint tokens; compressed attention may miss them. Predicted: most sensitive.

2 Method

2.1 GRC compression

Identical to Part I [1]: per-layer joint Gram eigendecomposition of $\mathbf{W}_Q^\top \mathbf{W}_Q + \mathbf{W}_K^\top \mathbf{W}_K + \mathbf{W}_V^\top \mathbf{W}_V$, projection onto leading-k eigenvectors, sign-canonicalised, calibration-free.

2.2 Benchmark protocols

MMLU 57 subjects, 5-shot prompting with subject-matched dev examples, answer extracted as `A/B/C/D` after "Answer:" token. Accuracy = fraction of correct multiple-choice answers.

GSM8K 1,319 test questions, 5-shot chain-of-thought prompting. Answer extracted as the number after "`####`". Correct if within 0.01 of ground truth.

HumanEval 164 Python programming problems. Pass@1: generate one completion, execute against test cases in a subprocess sandbox, pass if all tests succeed.

IFEval 541 instruction-following prompts with verifiable constraints. Strict accuracy: ALL constraints satisfied.

2.3 Compression ranks

$k \in \{256, 512, 768, 1024, 1536, \infty\}$, where $k{=}\infty$ is the uncompressed baseline. For Llama-3.1-8B (d=4096), these correspond to $k/d \in \{0.0625, 0.125, 0.1875, 0.25, 0.375, 1.0\}$.

3 Structural Predictions

Rationale. MMLU depends on FFN-stored knowledge; attention at k=512 (88.9% of d=576) still provides enough routing fidelity to select the correct answer from stored knowledge. GSM8K requires cross-position attention to intermediate computation tokens; on SmolLM2-135M, the model is too small to solve GSM8K at all (baseline 0%, regardless of compression).

3.1 Limitations of this study

This paper reports task-level measurements on SmolLM2-135M-Instruct (d=576, MHA). Measurements on larger models (Llama-3.1-8B at d=4096, GQA) are deferred pending access to $\geq$ 24 GB GPU hardware. The asymmetric degradation pattern (knowledge-retrieval resilient, reasoning fragile) is established at 135M scale; whether it generalizes to 8B-scale models with non-zero GSM8K baselines is an open empirical question.

4 Measured Results (SmolLM2-135M, Exp F1)

Protocol (Python/ChatML, 2026-05-02). We evaluated GRC-compressed SmolLM2-135M-Instruct (d=576, MHA, FP16) on MMLU (10 subjects, 2-shot, 30 questions/subject) and GSM8K (chain-of-thought, 2-shot, 50 samples) at $k \in \{256, 512, \text{full}\}$ via the Python transformers use (`scripts/task_bench_python.py`, RTX 4070 Laptop). Unlike the earlier `geodessical2`-based measurements, this harness uses the proper ChatML instruct template and a fresh model reload per rank to eliminate accumulated drift. HumanEval and IFEval are not measured in this revision.

Correction notice. The previous revision of this paper reported MMLU 28.1% at k=256 from the `geodessical2` binary without ChatML template support. The Python/ChatML remeasurement found 0.0% at k=256 — the earlier number was an artifact of plain-text prompting where the 135M model's few-shot formatting failure masked the compression collapse. We retract the k=256 row and replace all numbers below with the ChatML-gated remeasurement.

[†]GSM8K at full rank: 1/40 correct (2.5%) in a partial 5-shot run; 0/48 in the 2-shot ChatML run. SmolLM2-135M is too small for grade-school math regardless of compression.

Key findings.

Table 1: Measured task accuracy at GRC compression ranks on SmolLM2-135M-Instruct, Python/ChatML use (2026-05-02).

Benchmark	k=256	k=512	k=∞ (baseline)
MMLU	0.0%	23.7%	25.3%
GSM8K	0.0%	0.0%	~2.5%

1. Asymmetric degradation CONFIRMED. MMLU survives compression at k=512 (23.7% vs. baseline 25.3%, −1.6 pp)—knowledge recall depends primarily on FFN memory, not attention routing. At k=256 (k/d=0.44), MMLU collapses to 0%: attention routing is destroyed below the safe frontier.

2. Safe frontier at $k \geq 512$. For SmolLM2-135M (d=576), the safe compression rank for task performance is $k \geq 512$ ($k/d \geq 0.89$), matching the PPL-based safe frontier (Paper I, Exp A1). The model dimension is small enough that the attention subspace is intrinsically low-capacity; compressing below k=512 degrades both PPL and task accuracy.

3. GSM8K at 135M is not informative. The baseline model scores ≈0% on GSM8K, so we cannot measure compression-induced degradation. This experiment must be repeated on a model where GSM8K baseline >0% (e.g., Llama-3.1-8B) to quantify the reasoning-specific attention dependence.

4. Prior k=256 MMLU retracted. The 28.1% at k=256 reported in the earlier revision was an artifact of the `geodessical2` binary's lack of ChatML instruct template support. The Python/ChatML remeasurement corrects this to 0.0%.

Refinement of predictions. At SmolLM2-135M scale (d=576), the attention subspace is already at $k_{95}/d \approx 0.89$ (Paper I, Exp A1), so task performance does not degrade until k<512—but at those ranks PPL is already catastrophic. The practical message: task-level safety is gated by PPL safety for small models; if PPL is preserved, knowledge-recall task performance follows.

4.1 Safe Frontier Analysis (Exp F4)

For SmolLM2-135M, the safe compression frontier for task performance matches the PPL-based frontier ($k \geq 512$, Paper I). The model dimension d=576 is small enough that the attention subspace is intrinsically low-capacity: compressing below k=512 (k/d<0.89) degrades both PPL and task accuracy simultaneously.

5 Empirical Protocol

5.1 Done

- Benchmark use (`scripts/task_bench.py`) — complete. Supports MMLU (57 subjects, 5-shot), GSM8K (chain-of-thought, 5-shot), HumanEval (pass@1, subprocess sandbox).
- Exp F1 (SmolLM2-135M): MMLU and GSM8K measured at $k \in \{256, 512, \infty\}$ on EC2 L40S. Asymmetric degradation confirmed. See § 4.

- Exp F4 (Safe frontier analysis): For SmolLM2-135M, $k \geq 576$ (full dimension) is required for task safety—the same conclusion as the PPL-based safe frontier (Paper I, Exp A1). See `benchmarks/safe_frontier_analysis.json`.

5.2 Open

- GSM8K on Llama-8B (F2): Requires Llama-3.1-8B with ≥ 24 GB GPU. Not measured in this revision.
- HumanEval bimodal (F3): Requires HumanEval dataset and ≥ 24 GB GPU. Not measured in this revision.

6 Reproduction

```python
# Download datasets to data/
# MMLU: https://github.com/hendrycks/test
# GSM8K: https://github.com/openai/grade-school-math
# HumanEval: https://github.com/openai/human-eval

# Run full sweep (~6-8 hours)
python scripts/task_bench.py \
  --model models/llama3.1-8b-instruct-q4_k_m.gguf \
  --benchmark all --ranks 256,512,768,1024,1536 \
  --n-shot 5 --max-q 200

# Quick smoke test (10 questions each)
python scripts/task_bench.py \
  --model models/smollm2-135m-instruct-q8_0.gguf \
  --benchmark mmlu --ranks 256,1024 --n-shot 5 --max-q 10
```

7 Limitations

- All predictions in this revision are structural estimates, not measurements. They will be replaced once the use executes.
- Single model (Llama-3.1-8B). Cross-model task impact (SmolLM2, Gemma, Qwen) is deferred.
- Perplexity-only prior from Part I. The correlation between PPL degradation and task degradation is itself an empirical question that this paper will answer.
- Quantisation level (Q4_K_M) is fixed. Higher-precision weights (FP16) may shift the safe frontier.

References

[1] W. K. O. Stewart. "Geodesic Runtime Compression and the 106% Baseline Anomaly". In: *HyperTensor Part I, to appear* (2026).

[2] M. Geva, R. Schuster, J. Berant, and O. Levy. "Transformer Feed-Forward Layers Are Key–Value Memories". In: *EMNLP*. 2021.